

Subtracting Rational Numbers

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Rational numbers are numbers that can be written in the form

$$\frac{a}{b},$$

where a and b are integers and $b \neq 0$.

Same Denominator

For fractions with the same denominator, subtract the numerators and keep the denominator.

$$\frac{7}{10} - \frac{3}{10} = \frac{7 - 3}{10} = \frac{4}{10} = \frac{2}{5}$$

Different Denominators

For fractions with different denominators, first find a **common denominator**.

$$\frac{3}{4} - \frac{1}{8} = \frac{6}{8} - \frac{1}{8} = \frac{5}{8}$$

Examples

Example 1

Find $3.9 - 1.4$.

$$3.9 - 1.4 = 2.5$$

Example 2

Find $\frac{8}{3} - 2$.

$$2 = \frac{6}{3}$$

$$\frac{8}{3} - \frac{6}{3} = \frac{2}{3}$$

Example 3

Find $\frac{3}{4} - \frac{1}{8}$.

$$\frac{3}{4} = \frac{6}{8}$$

$$\frac{6}{8} - \frac{1}{8} = \frac{5}{8}$$

Practice Exercises

Practice 1

Find the difference.

$$3.9 - 1.4$$

Practice 2

Find the difference.

$$\frac{8}{3} - 2$$

Practice 3

Find the difference.

$$8.12 - 6.5$$

Practice 4

Find the difference.

$$2 - \frac{5}{4}$$

Practice 5

Find the difference.

$$\frac{7}{10} - \frac{3}{10}$$

Practice 6

Find the difference.

$$\frac{5}{6} - \frac{1}{3}$$

Practice 7

Find the difference.

$$\frac{3}{4} - \frac{1}{8}$$

Practice 8

Find the difference.

$$\frac{2}{5} - \frac{3}{5}$$

Practice 9

Find the difference.

$$\frac{7}{9} - \frac{2}{9}$$

Practice 10

Find the difference.

$$\frac{3}{4} - \frac{5}{8}$$

Activities

Activity 1

Solve the problem. Show your complete solution.

$$11.8 - 10.7$$

Activity 2

Solve the problem. Show your complete solution.

$$\frac{4}{3} - 1$$

Activity 3

Solve the problem. Show your complete solution.

$$3 - \frac{4}{3}$$

Activity 4

Solve the problem. Show your complete solution.

$$7.12 - 5.6$$

Activity 5

Solve the problem. Show your complete solution.

$$\frac{9}{12} - \frac{4}{12}$$

Activity 6

Solve the problem. Show your complete solution.

$$\frac{7}{8} - \frac{1}{4}$$

Activity 7

Solve the problem. Show your complete solution.

$$\frac{5}{6} - \frac{1}{3}$$

Activity 8

Solve the problem. Show your complete solution.

$$\frac{5}{2} - \frac{3}{2}$$

Activity 9

Solve the problem. Show your complete solution.

$$\frac{8}{11} - \frac{3}{11}$$

Activity 10

Solve the problem. Show your complete solution.

$$\frac{5}{6} - \frac{7}{12}$$

Thank you!